EVERY ADEQUATE FUNCTOR IS A QUOTIENT OF A DIRECT SUM OF LINEARLY ADEQUATE FUNCTORS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- An adequate functor embeds into a module-valued functor, and adequacy is detected on finitely presented algebras. Same chapter, Lemma lemma-adequate-finite-presentation, invoked at proof lines 477-479. Reduces the construction to finitely presented test algebras.
- Every module is a filtered colimit of finitely presented modules. Algebra, the standard filtered-colimit presentation of modules. Produces the finitely presented N' and the commutative square.

Source reference. Original lemma 06UU: sources/stacks-adequate/pinned-original-source.tex, source lines 410–410. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let F be an adequate functor on Alg_A. Then there exists a surjection $L \to F$ with L a direct sum of linearly adequate functors.*

Proof. Choose an exact sequence $0 \to F \to \underline{M} \to \underline{N}$ where $\underline{M} \to \underline{N}$ is given by $\varphi : M \to N$. By Lemma 06UU it suffices to construct $L \to F$ such that $L(B) \to F(B)$ is surjective for every finitely presented A-algebra B. Hence it suffices to construct, given a finitely presented A-algebra B and an element $\xi \in F(B)$ a map $L \to F$ with L linearly adequate such that ξ is in the image of $L(B) \to F(B)$. (Because there is a set worth of such pairs (B, ξ) up to isomorphism.)

To do this write $\sum_{i=1,\ldots,n} m_i \otimes b_i$ the image of ξ in $\underline{M}(B) = M \otimes_A B$. We know that $\sum \varphi(m_i) \otimes b_i = 0$ in $N \otimes_A B$. As N is a filtered colimit of finitely presented A-modules, we can find a finitely presented A-module N', a commutative diagram

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of A-modules

$$\begin{array}{ccc} A^{\oplus n} & \longrightarrow & N' \\ {\scriptstyle m_1,\ldots,m_n}\downarrow & & \downarrow \\ M & \longrightarrow & N \end{array}$$

such that $(b_1, \ldots, b_n)$ maps to zero in $N' \otimes_A B$. Choose a presentation $A^{\oplus l} \to A^{\oplus k} \to N' \to 0$. Choose a lift $A^{\oplus n} \to A^{\oplus k}$ of the map $A^{\oplus n} \to N'$ of the diagram. Then we see that there exist $(c_1, \ldots, c_l) \in B^{\oplus l}$ such that $(b_1, \ldots, b_n, c_1, \ldots, c_l)$ maps to zero in $B^{\oplus k}$ under the map $B^{\oplus n} \oplus B^{\oplus l} \to B^{\oplus k}$. Consider the commutative diagram

$$\begin{array}{ccc} A^{\oplus n} \oplus A^{\oplus l} & \longrightarrow & A^{\oplus k} \\ \downarrow & & \downarrow \\ M & \longrightarrow & N \end{array}$$

where the left vertical arrow is zero on the summand $A^{\oplus l}$. Then we see that L equal to the kernel of $\underline{A^{\oplus n+l}} \to \underline{A^{\oplus k}}$ works because the element $(b_1, \ldots, b_n, c_1, \ldots, c_l) \in L(B)$ maps to ξ. $\square$